Journal of Applied Geophysics 178 (2020) 104054

Contents lists available at ScienceDirect

Journal of Applied Geophysics

journal homepage: www.elsevier.com/locate/jappgeo

# Using deep learning based methods to classify salt bodies in seismic images

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ARTICLE INFO

*Article history:*
Received 5 September 2019
Received in revised form 24 February 2020
Accepted 4 May 2020
Available online 07 May 2020

*Keywords:*
Salt detection
Seismic
Segmentation
CNN

ABSTRACT

Identification of salt bodies buried under the earth's surface remains important for hydrocarbon industry as they form traps to hold oil and gas reservoirs in the subsurface. Seismic data interpretation on a large volume of data is labour-intensive and time-consuming task if performed manually. Many efforts have been made in the past to automate or semi-automate the process of extracting the salt bodies from migrated images using many proposed seismic attributes. There remains need of domain experts for picking the right geological features for interpretation and these attributes may not completely interpret the noise-contaminated seismic data. In the recent past, machine-learning algorithms have also been used to identify salt bodies using seismic attributes and they outperform the existing methods. However, there is still need to have directly assisted processes of seismic data interpretation without the compulsion of providing seismic features. Recent researches have used deep learning methods for seismic facies classification and fault detection, which resulted with better outcomes than other machine learning algorithms. In this document, we propose a novel deep learning method based on U-Net plus SeResNet to identify salt bodies in seismic images. This method does not require domain knowledge expert and it is expected to extract required useful features automatically from the seismic data. After 10-fold cross-validation, a promising intersection over union (IOU) score has been achieved.

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## 1. Introduction

Salt bodies are impermeable and form traps where natural gas and petroleum can be collected as shown in Fig. 1-a. Seismic imaging is used to capture images of earth's subsurface for oil and gas exploration. This method uses sound waves, which penetrate inside the earth's surface and are recorded using geophones when they reflect from the subsurfaces. It gives a lot of information including the depth, density and thickness of the subsurface rocks. It is like an ultra-sound of earths subsurface. A seismic migrated image is shown in Fig. 1-b where a salt body can be located in middle.

Seismic survey data has now entered 'big data' category and this has become a challenge to obtain analytical value of already acquired data when huge amount of new data is coming in different forms and shapes (Roden, 2016). A lot of data remains unutilized because it is not possible to be analyzed manually in limited amount of time. To localize salt bodies or hydrocarbon in seismic data it is essentially required to automate the process to achieve higher degree of reliability. Some semi-automated tools have been introduced to assist the domain experts, but data explosion demands domain experts' knowledge to be formalized and incorporated in new or the existing tools to reduce man-power and valuable time.

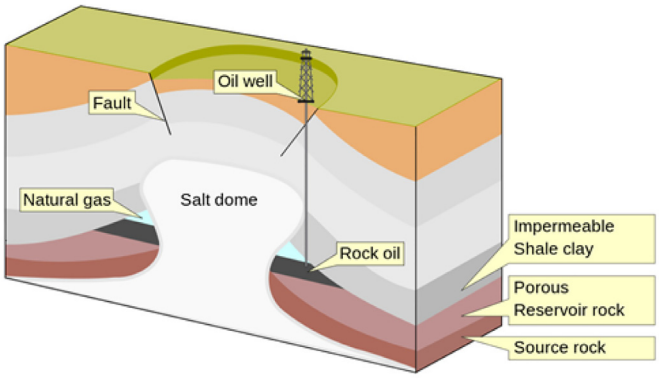
Seismic interpretation is based on seismic attributes which are measurable properties of seismic data. There has been an explosion in the quantity of seismic attributes and new ones are added frequently with the passage of time (Chopra and Marfurt, 2005). As quantity of attributes increases, their relationship and lack of clarity between their interdependencies cause more complexity for even domain experts to grasp all at once. Earlier studies have focused on finding salt body attributes in the seismic migrated images, which is approximately the same.

Machine learning based multi-attribute solutions have also been proposed to extract the useful features to automate the process of classification of salt bodies in seismic images. For this purpose, both supervised and unsupervised techniques have been used including combination of some classifiers in ensemble-based methods too (Guillen et al., 2015; Di et al., 2017).

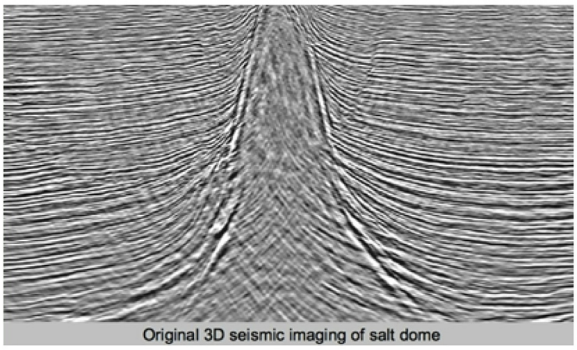
Recently deep learning based methods have outperformed the conventional machine learning algorithms in several fields i.e. computer vision (Voulodimos et al., 2018). So far, this technology has not gained any major breakthrough within oil and gas industry. In contrast to seismic-attributes based methods, deep learning algorithms work directly on the seismic data, learn the most

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https://doi.org/10.1016/j.jappgeo.2020.104054
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a: Salt bodies in subsurface [1]




b: Seismic migrated image [2]

**Fig. 1.** a: Salt bodies in subsurface (https://www.nap.edu/read/25032/chapter/13, n.d.). b: Seismic migrated image (https://www.hillgeo.com/salt-dome-imaging/, n.d.).

discriminative features on its own, and does not require any additional attributes. Therefore, we propose this method to automatically classify and outline the salt bodies in seismic data.

## 2. Literature review

Salt identification in seismic images has been previously addressed in the literature under the titles of salt body detection, salt-dome detection and salt-boundary detection. In literature, it is considered as a two-step problem, computing the salt attribute image and then detecting the salt boundaries from that attribute image. Salt boundary detection has been considered as segmentation problem. Previous studies have mainly focused on finding new seismic attributes for salt detection. In this section, we will cover major salt attributes, techniques based on these attributes, machine learning based techniques and use of deep learning for processing seismic data.

**Table 1**
Summary of recent deep learning methods.

| Year | Architecture | Evaluation Metric | Dataset | Reference |
|---|---|---|---|---|
| 2018 | Seg-Net | Accuracy | SEAM | Shi et al., 2018 |
| 2018 | Simple CNN | – | – | Waldeland et al., 2018 |
| 2019 | encoder-decoder CNN | Accuracy, F1 Score | SEAM | Shi et al., 2019 |
| 2019 | U-Net + Res-Net | IOU | SEAM | Zeng et al., 2019 |
| 2019 | U-Net | IOU | N/A | Wang et al., 2018 |

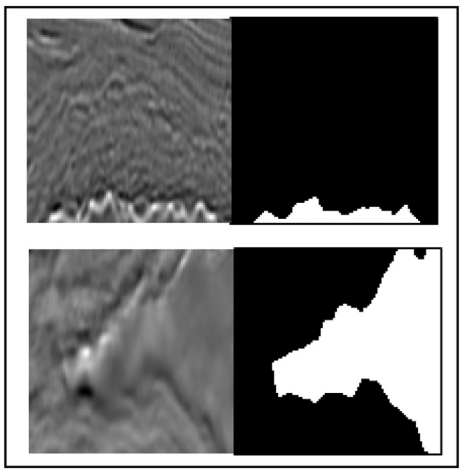


**Fig. 2.** Seismic image and classified salt region.

### 2.1. Segmentation based Techniques

Beginning with the simplest edge-detection based method that was first used in 2007, when a canny edge detector was used to detect salt boundaries in post stack seismic data (Jing et al., 2007). A 3D multidimensional edge detector was used to detect salt dome in migrated seismic data of Netherlands offshore F3 block (Asjad and Deriche, 2015b). Further mentionable work is combining the edge-based techniques with dip guiding (Aqrawi et al., 2011) and texture-based attributes (Deriche, 2016) to get better results than using them independently.

### 2.2. Seismic attribute based methods

Texture based attributes (Berthelot et al., 2013; Hegazy and Al Regib, 2014), reflection dip and frequency attributes (Halpert and Clapp, 2008), seismic disorder attributes (Chopra and Marfurt, 2016), salt likelihood (Wu, 2016) and saliency-based attribute (Shafiq et al., 2016) are used for highlighting salt boundaries. Combination of Gray Level Co-occurrence Matrix (GLCM) attributes with Gabor filters have turned out to be work better than other methods for delineating salt boundaries in seismic data (Amin et al., 2015; Amin and Deriche, 2016). Segmentation techniques based on normalized cut image (Lomask et al., 2007), modified PRC algorithm (Halpert et al., 2014) and sparse representation theory (Ramirez et al., 2015) are used for detecting salt dome boundaries. Although these automated methods assist in detection of salt bodies, but domain expert's intervention is still required in complex structures to get accurate results.

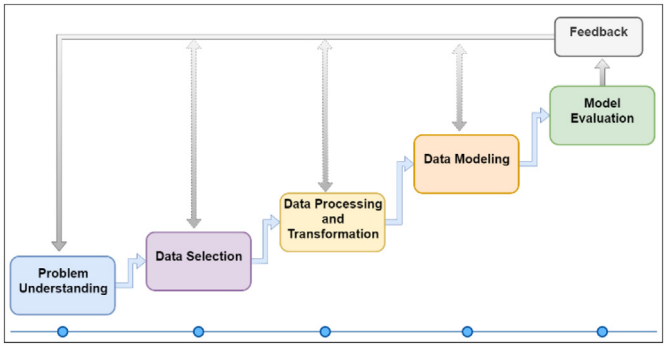



**Fig. 3.** Methodology.

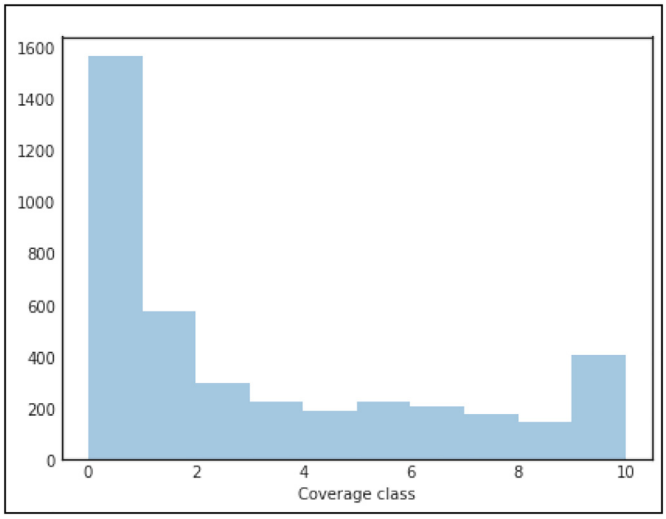



**Fig. 4.** Class coverage.

### 2.3. Mahine learning based methods

Machine learning methods have been applied in seismic data analysis for fault detection (Zhang et al., 2014) and facies classification (Zhao et al., 2015). Salt boundary detection problem met machine learning in 2015 when Extremely Randomized Trees Ensemble algorithm (Geurts et al., 2006) was used to automatically classify salt regions using SEAM I (SEG Advanced Modeling Corporation) dataset. Attributes used for classification were signal amplitude, second derivative, and curve length (Guillen et al., 2015). Further, a K-Means clustering technique is used for highlighting salt boundary in F3 seismic dataset over the Netherlands North Sea and twelve seismic attributes were used for this purpose (Di et al., 2017).

### 2.4. Limitations of conventional methods

Use of seismic-attribute based methods need professional assistance, intuition and judgment, which can lead to errors. There a lot of attributes are proposed and still being added. It is difficult and burdensome to study their significance and interdependency. Conventional methods need to be repeated when shifting the exploration from one dataset to another, considering the distinct geology in different areas, which is again an overhead.

### 2.5. Deep learning based methods

Traditional methods use a cumbersome trial and error process of feature extraction which requires expert's judgment in choosing the best features in the data. Unlike traditional approach, deep-learning based methods use end-to-end learning process which only requires dataset of images with annotate classes and then underlying patterns and features are automatically discovered.

Recently deep learning methods have been used in seismic data analysis for fault detection (Araya et al., 2017) and seismic facies classification (Salles Chevitarese et al., 2018) and they came up with promising results. Few studies have been carried out for salt detection problem as well on different datasets. Below is the brief overview of the recent methods used for salt classification problem and they are also summarized in Table 1.

A research (Shi et al., 2018) published in 2017 used a seven layered CNN architecture based on encoder-decoder model of SegNet (Badrinarayanan et al., 2017) for salt classification using SEAM Phase 1 dataset and results interpreted reflected good performance for classifying salt bodies. Another recent study has discussed how CNN can be useful in delineation of salt bodies in 3D seismic volumes (Waldeland et al., 2018). An encoder-decoder based network used for segmenting salt bodies in 3D seismic volumes shows good results on SEAM Phase 1 dataset (Shi et al., 2019). In a recent work in 2019, U-Net has been used with ResNet blocks as model architecture and intersection over union (IOU) is used as scoring metric (Zeng et al., 2019; Wang et al., 2018). In this paper, we will modify this architecture using SE-ResNet.

## 3. Methodology

Conventional algorithms for body detection depend on seismic attributes even machine learning based solutions use these attributes for training the ML model. This involves professional assistance, intuition and judgment, which can lead to errors and biasness in some cases. Labor intensiveness and time consumption are major issues. Deep learning based methods, on the other hand, do not require any such guidance rather they search for the most discriminating features in the raw images data by their own. This property has made them so popular that they are extensively being used in computer vision problem and achieving a very good performance over the conventional methods provided a lot of data for training.

In this research we have taken this as semantic segmentation problem in which input images are classified pixel-by-pixel into salt and non-salt regions. In the corresponding mask against an image, salt regions are represented by white pixels while non-salt regions are in black as shown in Fig. 2.

### 3.1. Dataset

This dataset is provided by a Geophysical Company TGS and available for public at kaggle (https://www.kaggle.com/datasets, n.d.). The data contain seismic images at various locations chosen at random in the subsurface.

The image dimensions are 101 × 101 pixels and each training image has a corresponding mask for the salt body region indicating the pixels classified as salt. Fig. 2 shows seismic images with corresponding salt body masks indicating the region identified as salt. Along with this, we also have the sub-surface depth for all images that turns out to be useful in training process.

### 3.2. Workflow

Basic overview of the methodology is shown in Fig. 3 and individual modules will be discussed in detail in implementation section. Data was

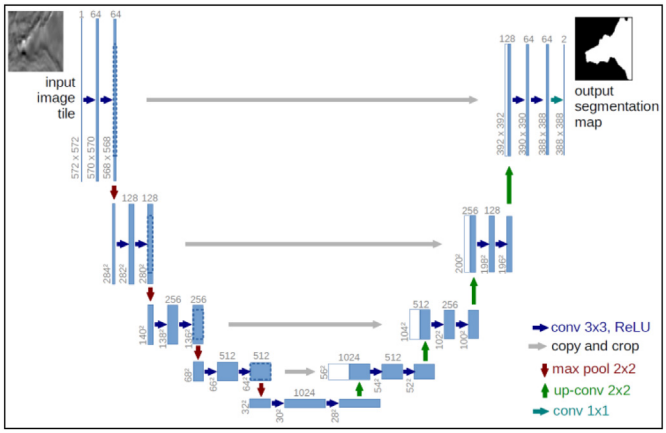



**Fig. 5.** U-Net architecture.

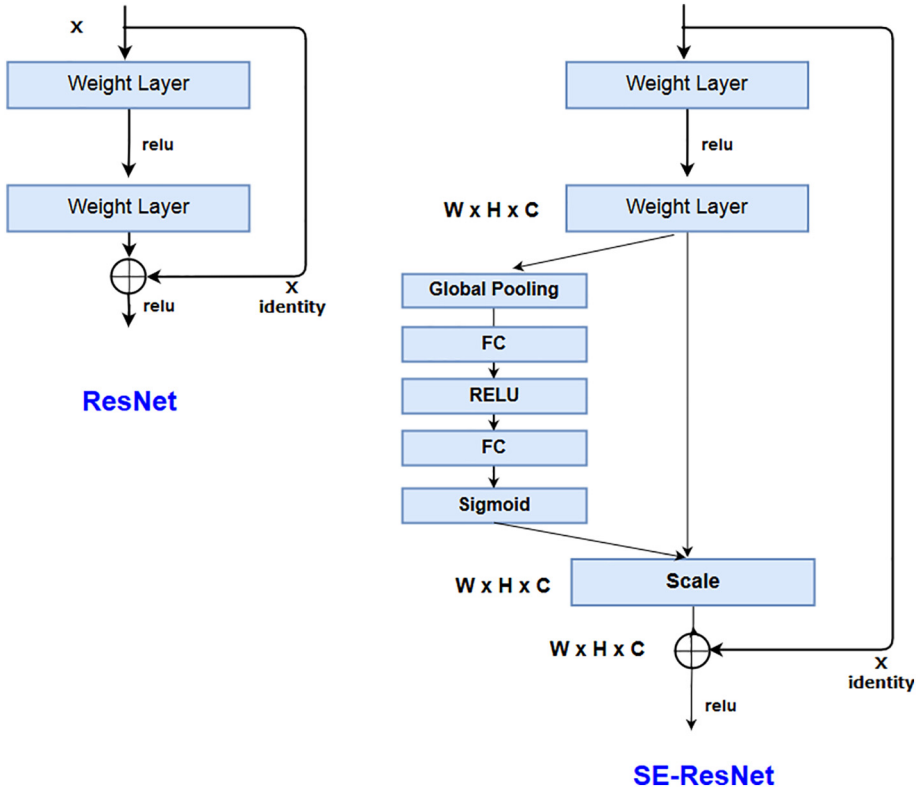



**Fig. 6.** Squeeze and excitation networks.

first prepared to bring in a form that can be fed to a Machine Learning algorithm. Our CNN model architecture is comprising of base U-Net architecture with simple modified ResNet blocks with addition of Squeeze and Excitation Networks referred as SE-ResNet. Model was then trained, first using combination of binary cross entropy (BCE) and dice loss and then using Lovasz Loss. The model was evaluated using k-fold cross validation. Evaluation metric for this segmentation problem was intersection over union IOU.

As it can be observed from Fig. 3 that we have iterative steps in the work flow. We keep on transforming the data until it fits into the data model. We also make changes in the network architecture with respect to the model evaluation results. These changes may require re-training of the model.

## 4. Implementation

### 4.1. Data preparation and loading

Data used in this study is 101 × 101 pixel images and their corresponding masks of the same dimension in separate folder with the same name. Images are taken as training data and masks are the training labels in this case. Images are read from the directory and are converted to grayscale first. Image dimensions are by default (101 × 101 × 3). The image is reduced to (101 × 101 × 1) to be fed into neural network. After loading the images, they are converted to *Numpy Array* to make computation more flexible. Image array is normalized to have values between 0.0 and 1.0. Similar procedure is carried out for image masks as well.

**Table 2**
Learning rate scheduling.

| Epochs | Loss function | Optimizer | Learning rate |
|---|---|---|---|
| 0–300 | BCE + Dice loss | Adam | 0.01 |
| 301–600 | Lovasz | Adam | 0.001 |

### 4.2. Stratified training and validation split

Available data is partitioned to training and validation set using stratified split based on salt coverage on masked images. Class coverage is calculated from salt coverage and used as a new column appended to the dataset. Fig. 4 shows the class coverage on the dataset.

$$\text{Coverage} = \frac{Sum\ of\ \ mask\ pixels}{(Image\ Size)^2} \tag{1}$$

**Table 3**
k-fold validation results.

| Fold | IOU Metric score | | | |
|---|---|---|---|---|
| | BCE-Train | BCE-Valid | Lovasz-Train | Lovasz-Valid |
| 1 | 0.8682 | 0.8193 | 0.9083 | 0.853 |
| 2 | 0.8706 | 0.822 | 0.8939 | 0.835 |
| 3 | 0.8686 | 0.8255 | 0.9079 | 0.851 |
| 4 | 0.861 | 0.807 | 0.8949 | 0.834 |
| 5 | 0.8541 | 0.8223 | 0.9005 | 0.841 |
| 6 | 0.8751 | 0.8197 | 0.9101 | 0.842 |
| 7 | 0.8698 | 0.8233 | 0.9103 | 0.837 |
| 8 | 0.8512 | 0.8195 | 0.9102 | 0.85 |
| 9 | 0.8756 | 0.823 | 0.9081 | 0.835 |
| 10 | 0.8624 | 0.815 | 0.9101 | 0.8421 |
| Avg | 0.866771 | 0.819871 | 0.9037 | 0.84201 |

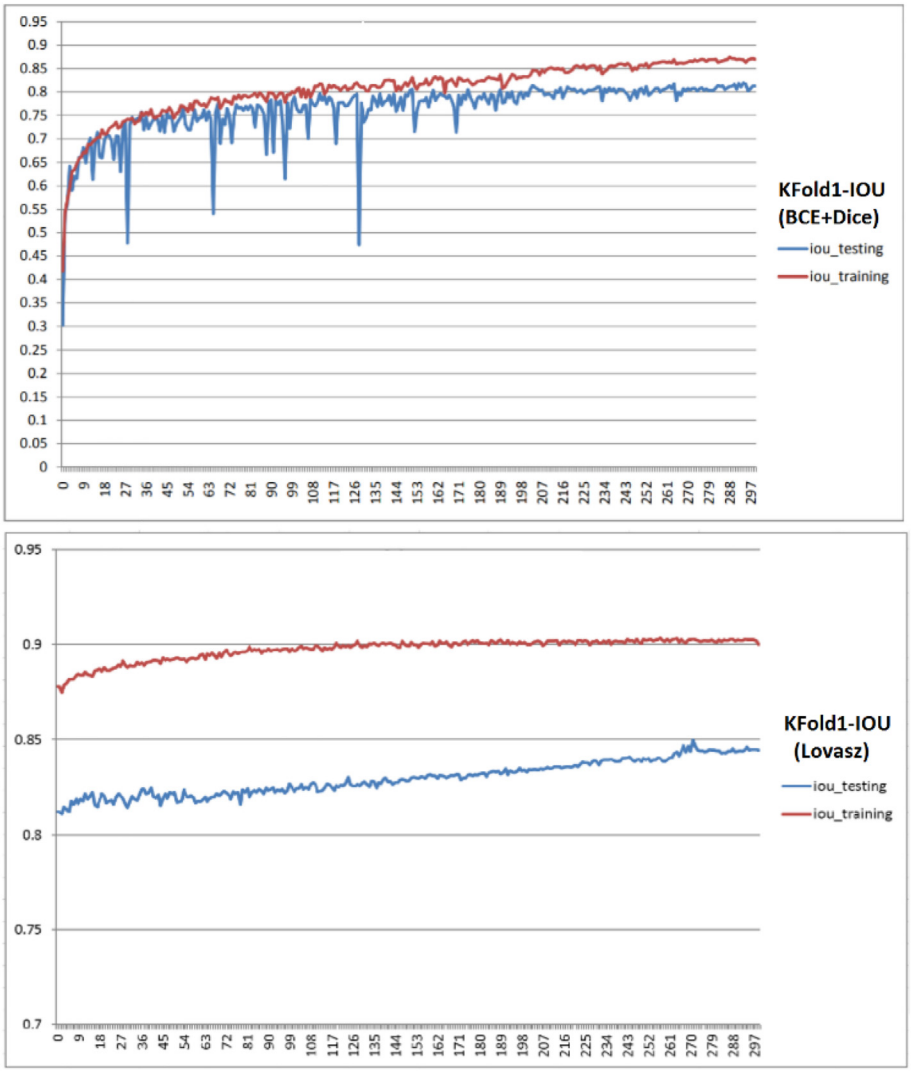



**Fig. 7.** a: K-Fold1 IOU score with BCE Vs Lovasz Loss. b: K-Fold9 IOU score with BCE Vs Lovasz Loss.

### 4.3. Model architecture

In this study, simple U-Net architecture was used with addition of modified ResNet blocks. U-Net (Ronneberger et al., 2015) was developed mainly for biomedical image segmentation problem in 2015 but it has performed exceptionally for other segmentation problems as well. Segmentation is image processing approach that is used to partition image into different regions. Basic U-Net architecture is described in Fig. 5. There are two major parts i.e. down sampling also known as encoder and up-sampling called decoder. In encoder part, size of image contracts while depth increases. This part is responsible for learning global characteristics. In decoder or up sampling, image size slowly increases back while depth decreases. This part is responsible for learning the local characteristics in the image data.

This U-Net was embedded with modified simple ResNet blocks (He et al., 2016). Residual networks were proposed in 2015 for degradation problem in deeper networks on certain point due to over-fitting. It was overcome by skipping the training of some layers after adding skip connections also called *identity shortcuts* as shown in Fig. 6.

A modification to these residual blocks known as SE-Resnet was introduced in 2017 for improvements in channel interdependence. It also improved the results of ImageNet competition by 25% with respect to previous year (Hu et al., 2017). It is basically a two-step process that reduces the computation to a good extent and help to model to converge in a smaller number of iterations. Using global average pooling each channel is squeezed to a numeric value first. A dense layer followed by a non-linearity is added and its complexity is lowered by a certain ratio then is fed into another dense layer followed by a non-linearity i.e. sigmoid. The output values are then used to weigh each feature's map of Convolutional block as the last step that scales each channel based on its significance.

### 4.4. Model evaluation metric

CNN models are evaluated differently based on the problem and model architecture. Accuracy, Precision and Recall and mostly used as evaluation metrics. For this Segmentation problem, we have used Intersection over union IOU as evaluation metric. This method needs original bounding boxes from the labelled training set and the predicted

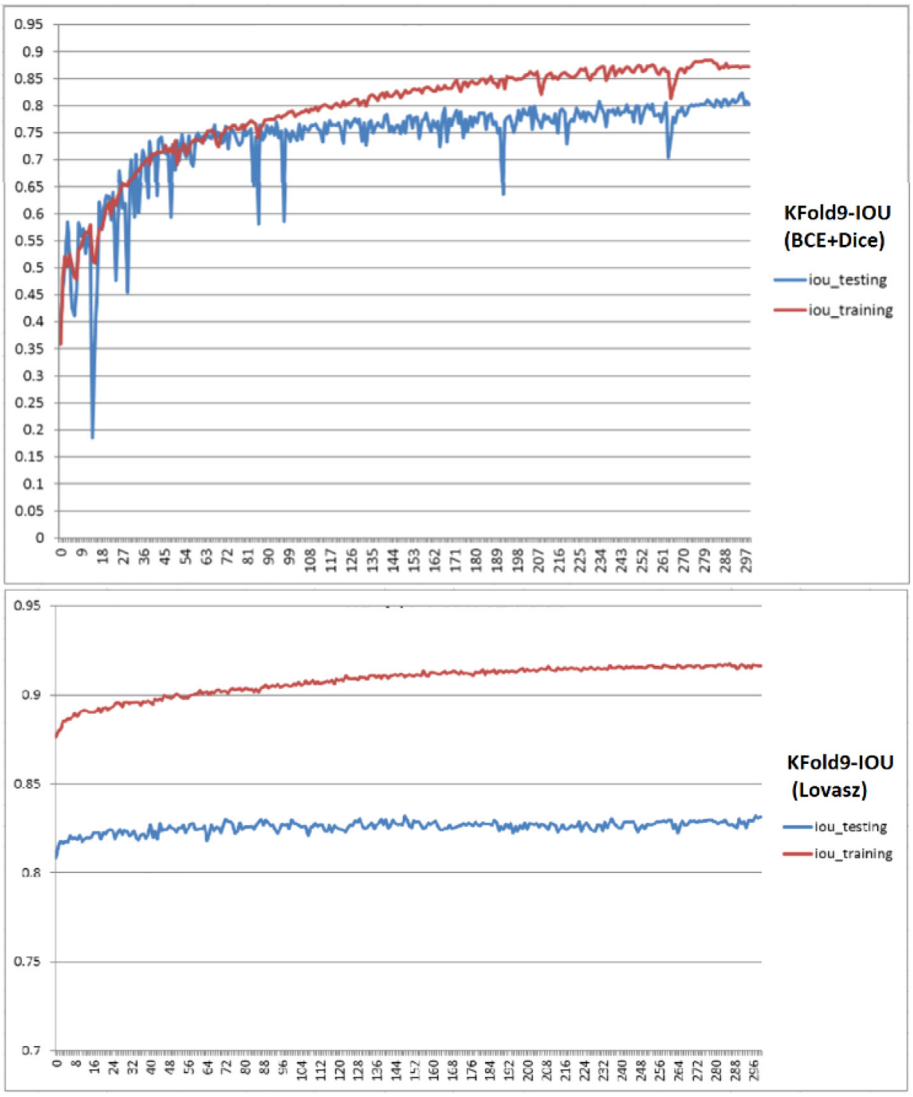



**Fig. 7** (*continued*).

bounding boxes from the trained model. It simply divides the area of overlap for these two bounding boxes by the total accumulated area as shown in Eq. (2). An IOU score greater than 0.5 is generally considered as a decent prediction.

$$\text{IOU} = \frac{Area\ of\ Overlap}{Area\ of\ Union} \tag{2}$$

### 4.5. Model training

This model was trained for about total of 600 epochs in two steps using GPU based environment provided by Google Colab. Loss function and learning rate were modified after first 300 epochs. Further details have been discussed in sub-section below.

### 4.6. Tools and technology

We used keras, a Tensor flow wrapper for creating network architecture and numpy library for data preparation and loading. Code and data were then uploaded to Google Colab for training the model (https://colab.research.google.com, n.d.). It is a Jupyter Notebook based GPU environment enriched with all required python libraries for data preparation and modeling. It provides 12 GB of Memory and NVIDIA Tesla K80 GPU that can be used for 12 h continuously if the allocated memory and disk space is not completely utilized.

### 4.7. Activation functions

For our network to learn complicated curves we need to add a non-linearity to it called activation function. ReLU proposed in 2010 (Nair and E. Hinton, 2010) has solved the vanishing gradient problem faced with non-linearity like sigmoid and tanh but suffered with dying ReLU problem as the gradient for negative values would be zero causing the neurons to die and no weight adjustment will occur. A modified version of ReLU called *Leaky ReLU* was proposed in 2015 that solved this issue by adding a small non-zero constant to the equation. Exponential Linear Units (ELU) (Clevert et al., 2015) proposed in 2015 turned out to be a strong alternative to ReLU. As we can see in Eq. (3), for non-negative values it is identity function like ReLU but unlike ReLU, it gets smooth slowly until its output reaches negative $\alpha$. We have used this activation function in hidden layer of our network.

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha.(e^x - 1), & x \le 0 \end{cases} \tag{3}$$

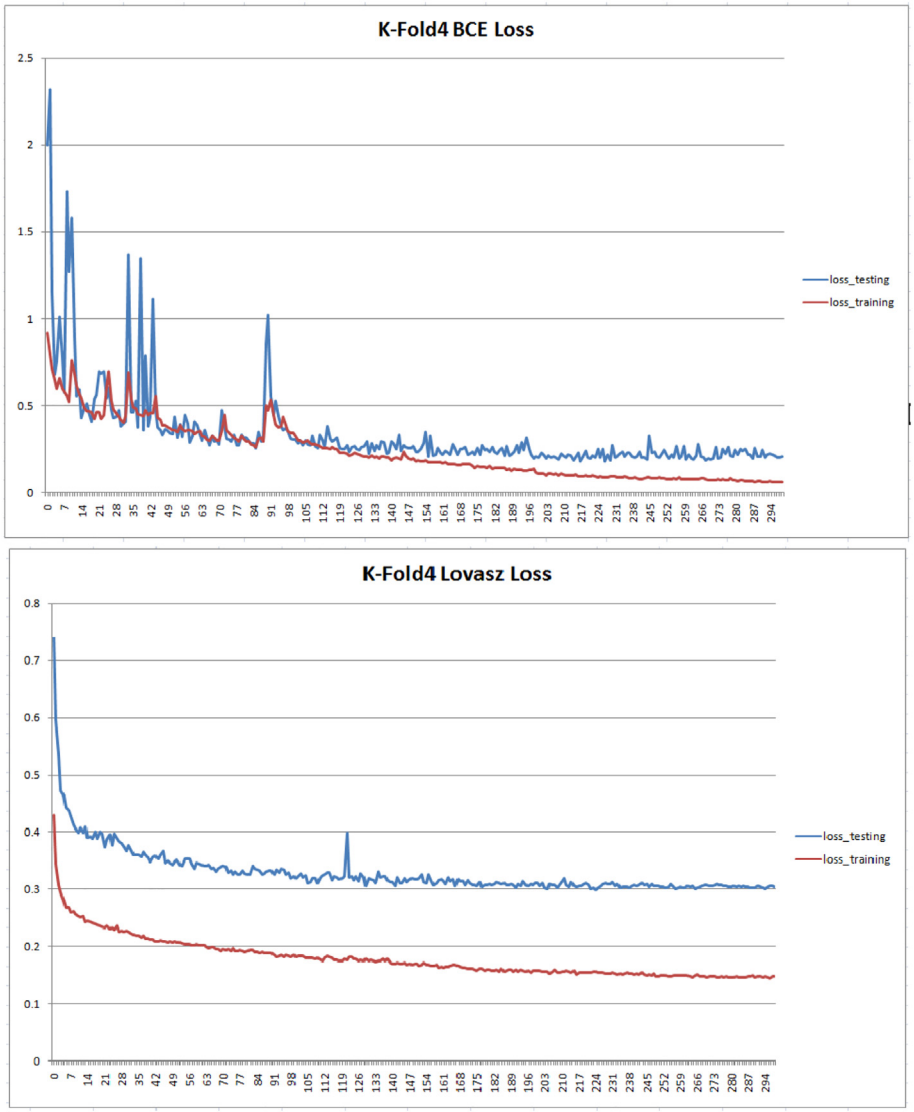



**Fig. 8.** K-Fold4 BCE Vs Lovasz loss decay.

### 4.8. Loss functions

This is a binary classification problem, as we have to classify an image pixel by pixel to check whether it is salt or not. Generally, Binary Cross Entropy is used as binary loss function in these types of problems. We have another popular function based on Dice Coefficient called Dice Loss that measures the overlap between data instances as mentioned in Eq. (4).

$$\text{Dice} = \frac{2\ |A \cap B|}{|A| + |B|} \tag{4}$$

We have combined BCE and Dice Loss, resulting a hybrid loss function, and used it for training the model to first 300 epochs. For fine-tuning and optimization of IOU metric, Lovász-Softmax loss turns out to be a better function (Berman et al., 2018).We used this loss for training the model for next 300 epochs. It is slower than other loss functions.

### 4.9. Learning rate and optimizers

Learning rate was scheduled as given in Table 2. In addition to that, we also used *ReduceLROnPlateau* policy to reduce learning rate by factor of 0.5 after no improvement in validation metric for 40 consecutive epochs.

## 5. Results

We evaluated the model using 10 k-fold cross validation with 600 epochs for each fold. Training comprises of two steps: For first 300 epochs model was trained using combination of BCE and Dice loss and for next 300 epochs it was further trained using Lovasz Loss function. To apply Lovasz Loss function we need to remove the last "sigmoid" activation. In addition, IOU metric needs to be modified as the default threshold for pixel prediction is 0 rather than 0.5. Training and validation IOU score for all folds are listed in Table 3 separately for both loss functions. At the end, average IOU score is calculated from them. Cross validation, results are discussed in detail in the following section.

## 6. Discussion

We have shown Training/Validation plots for fold-1 and fold-9 in Fig. 7a and b respectively. First graph in figure shows the IOU improvement over the number of epochs using combination of BCE and Dice Loss. From these graphs, we can see improvement in the model using

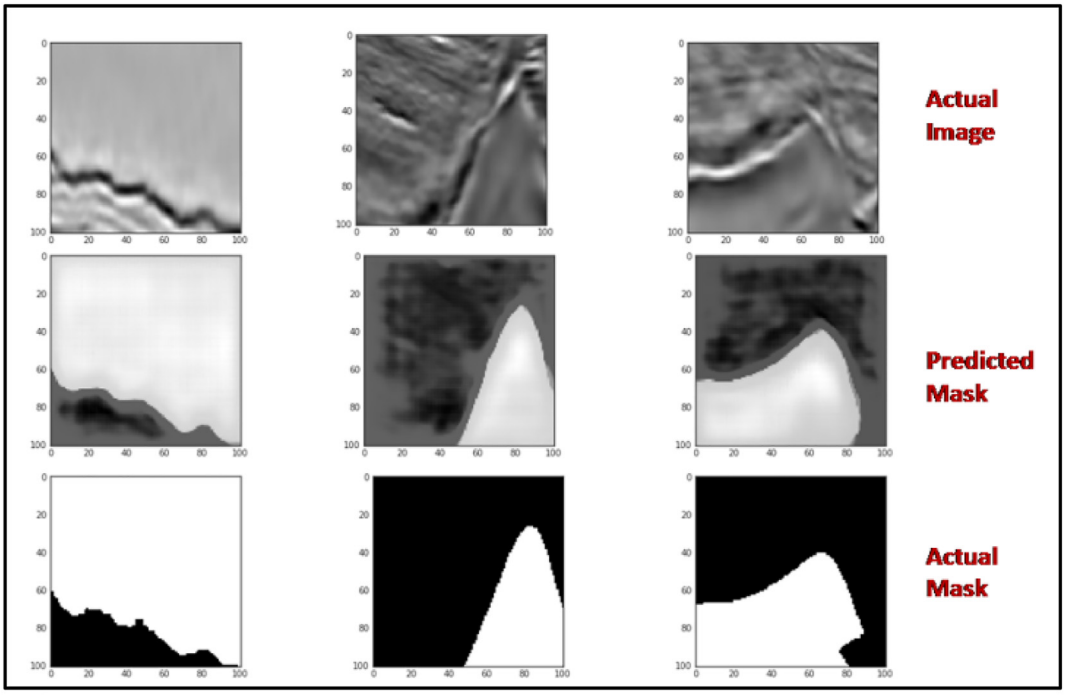



**Fig. 9.** Validation results.

BCE and loss is negligible after certain time. We need to change the loss function for further improvement and optimization of IOU metric. Second graph shows the IOU improvement after using Lovasz Loss function. From Fig. 7 and Table 2 we can see that Lovasz Loss function has increased IOU score on average value of 2%. We have also plotted the loss decay using both the loss functions in Fig. 8.

Segmentation results on unseen validation data are shown in Fig. 9. Fist row contains the actual images in grayscale from validation data. Second row represents the predicted masked from trained deep learning model before thresh-holding. And last row contains the actual masks to compare the performance against the predicted ones. Here we can see that the predicted results on validation data are promising and U-Net based deep learning model has classified the salt regions correctly.

## 7. Conclusion

In this paper, we have proposed a deep-learning based scheme for end-to-end automatic salt body segmentation in seismic images. Unlike conventional methods, this method does not require hand-crafted seismic features obtained through a trial and error process with expert's assistance. We have used U-Net architecture with modified ResNet blocks and evaluated the model using K-Fold cross validation. A decent intersection over union (IOU) verifies the capability of the skip-connection based CNN architecture for learning essential seismic features and efficiently generalizing training as well as validation dataset.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

**Muhammad Saif ul Islam:** Conceptualization, Methodology, Validation, Visualization, Writing – original draft. **Zeeshan Ahmed:** Supervision. **Muhammad Usama:** Writing – review & editing.

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